

COMSM0127 Immersive Interaction and Audio Design

Week 3, Lab B: Sound Design and Dynamic Audio

Dr Richard Hemming, Chris Bevan and Pete Bennett

Today's Lab

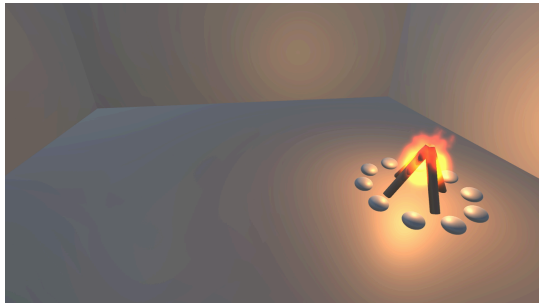
Last week you put a spatialised sound in a room. Today you **make** a sound, and make it change.

You heard the mystery sound at the start. Rebuild it two ways:

- **Part A, Recorded:** record it, trim it, and play it back so it never repeats.
- **Part B, Procedural:** sculpt it from noise in code, and make it respond to the game.

	09:25 – 09:50	**09:55 – 10:20**
Groups 1, 2	**Part A: Recorded**	Part B: Procedural
Groups 3, 4	**Part B: Procedural**	Part A: Recorded

Recording outside? Be back in the room by 09:35 (first half) or 10:10 (second half). At **09:50** everyone stops for five minutes, then swaps at 09:55. At **10:25** everyone comes back together to listen.



Goals to Complete Before the End of the Lab

By the end of this lab you should have:

1. Listened to the mystery sound and named the parts it is made of.
2. **Part A:** recorded those parts and played them back so no two plays are the same.
3. **Part B:** sculpted those parts from noise and made one respond to the fire's Intensity.
4. Added **one line of C#** in each part.
5. Played your fire to the room.

Basic goal: Campfire A and Campfire B both working. You leave with a complete Part A and a complete Part B in one project.
Stretch goal: one fire with recorded and procedural layers together.

Opening the project

1. **github.com/RichSF1210/W3B_FireLab**: green **Code** button, **Download ZIP**.
2. **Unblock the zip before extracting.** Right-click it, **Properties**, tick **Unblock**, **OK**.
3. Right-click the zip, **Extract all** (Desktop is fine).
4. **Open the scene directly:** in File Explorer, `Assets/W3B_Fire/Scenes`, double-click **W3B_Student_Shell**. Unity (**6000.3.19f1**) opens the project, first time in a few minutes. No Unity Hub needed. **This is the scene you work in.**
5. Press Play. Walk with **W A S D**; hold the right mouse button to look.

In the project	What it is
`W3B_Student_Shell`	**You start here.** A room, and two silent campfires: **Campfire A (Recorded)** for Part A, **Campfire B (Procedural)** for Part B
`W3B_Complete_Reference`	Both campfires finished. Look at it later to compare, not before you try
`W3B_Demo`, `W3B_Finale`	The lecturer's demos, to replay later

Each campfire has three empty layers, **Layer 1, 2, 3**, each with a spatialised Audio Source as in Week 2 Lab B: **one part of the sound per layer.** Save the scene (Ctrl+S) as you go.

First: listen

Headphones on, eyes closed. Listen to the mystery sound again: `Assets/W3B_Fire/Audio/Mystery`.

As a group, agree:

- How many separate parts can you hear?
- Describe each one: high or low? Constant, or in short bursts?
- Which part tells you most strongly what the sound is?

Give each part a name of your own and write it down. **One part per layer**, on both campfires.

A

Part A: Recorded

A1. Record (about 10 minutes)

Take a Zoom recorder outside the lab. For each part on your list, find something that makes a similar sound: scrunch, snap, spray, blow, rub, tear.

- Get close. Keep it quiet around you.
- Record **several takes** of each part. For short sounds, aim for **at least four** different ones.
- About 10 seconds per take is plenty. Say the part's name before each take, so you can find it later.

Safety: do not record people without asking. **Back in the room by 09:35 (first half) or 10:10 (second half).**

A2. Get the files off and tidy them (about 10 minutes)

1. **Off the recorder:** plug the Zoom into a lab PC with its USB cable (choose *Card Reader* on its screen), or take out the SD card. Copy the WAV files to your desktop.
2. **Audacity:** open a file. Select the part you want, **Edit > Remove Special > Trim Audio** (or Ctrl+T).
3. **Short sounds:** one sound per file, starting right on the sound.
4. **Constant sounds:** a few seconds that loop without a click (cut where the waveform crosses the centre line).
5. **File > Export Audio**, as **WAV**. Name it clearly, with no spaces: `snap_01.wav`, `snap_02.wav`, `air_loop.wav`.

Drag the WAVs into Unity's Project window, into `Assets/MyAudio`. Unity accepts WAV, AIFF, MP3 and OGG. **The names do not matter to the scripts.**

A3. Play them back

Work on **Campfire A (Recorded)**.

Constant parts (a loop): select one of its layers, **Add Component > SampleLoopLayer**, drag your loop into the Audio Source's **AudioClip**.

Short parts (several files): select a layer, **Add Component > MyRandomPlayer**. Lock the Inspector (padlock, top right), drag your files onto **Clips**, then unlock.

Press Play and listen. Then listen again. **What gives it away as a recording?**



A4. Your one line: make it vary

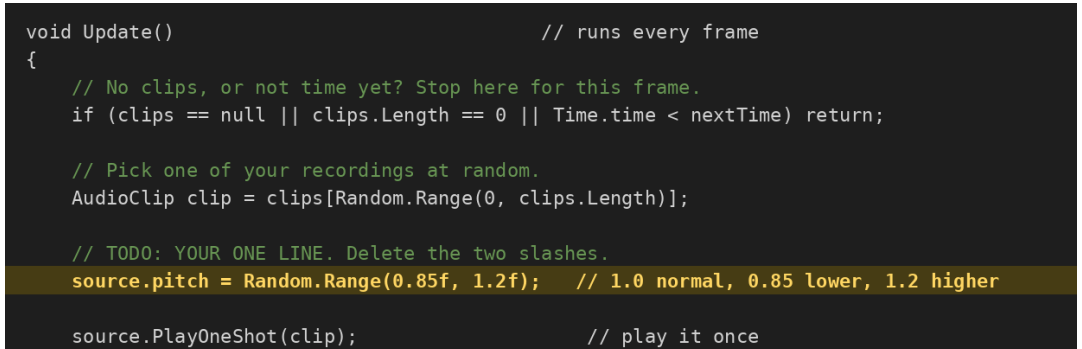
Double-click the **Script** field on My Random Player. Find `TODO`. The line is already there: **delete the two slashes** in front of it.

```
// TODO: YOUR ONE LINE. Delete the two slashes at the start of the next line.
source.pitch = Random.Range(0.85f, 1.2f);
//   Sets the playback pitch to a random number between 0.85 and 1.2.
//   1.0 is normal; 0.85 is lower and slower; 1.2 is higher and faster.
```

Where the shape comes from: in a recording it is baked in. Code can only change how it is played back: which file, when, how loud, what pitch. That is the limit of Part A.

Save (Ctrl+S). Wait for Unity's spinner. Press Play.

The files never change. The playback never repeats. This is dynamic audio from recordings.



B

Part B: Procedural

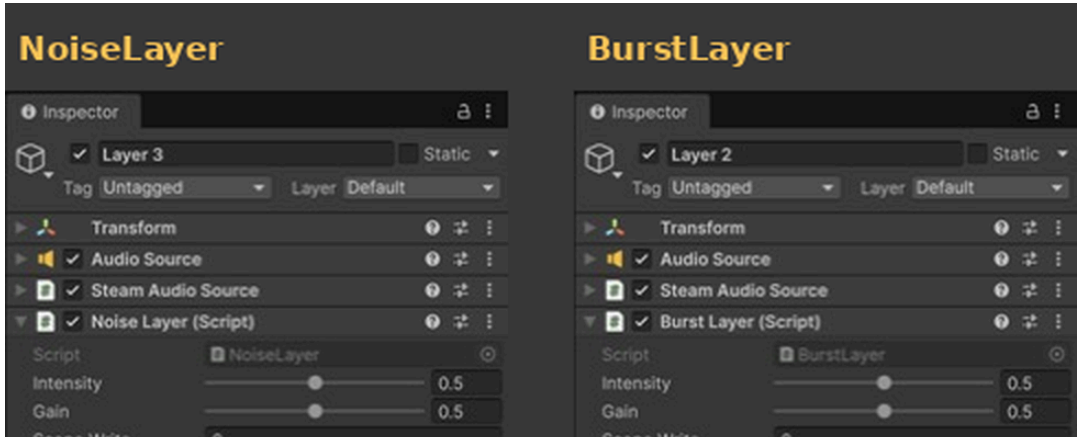
B1. Two building blocks

No audio files: these make sound from numbers, as it plays.

Work on **Campfire B (Procedural)**. Every procedural sound gets its shape from four steps:

Step	What it decides	NoiseLayer	BurstLayer
Source	Raw material	White noise	White noise
Filter	What it seems made of	Cutoff: lows = roar, highs = hiss	Brightness: high = small, hard things
Envelope	How it moves in time	A slow wobble: it breathes	Instant attack, short decay: a snap
Randomness	That it never repeats	Random wobble	Random timing and size

Select a layer, **Add Component**, and add one. Each script explains every line inside it.



B2. Sculpt by ear

Press Play. Move the sliders **while you listen**.

- NoiseLayer, Low Pass, low cutoff: what does it remind you of? High Pass, high cutoff?
- BurstLayer, fast, short and bright: what is it? Slow, long and dark?

Match each layer to one part on your list. Set the other layers' **Level** to 0 while you tune one.

Sliders changed in Play mode reset when you stop. Write your best values down, or right-click the component: *Copy Component*, stop, then *Paste Component Values*.

B3. Your one line: make it respond

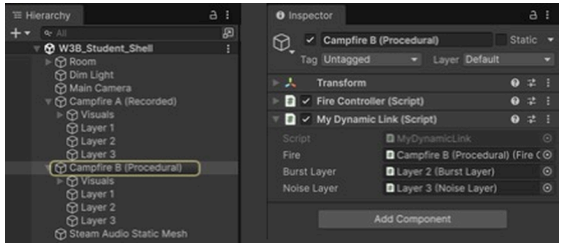
Campfire B has a **FireController** with an **Intensity** slider: 0 is embers, 1 is a blaze.

1. Select **Campfire B**, **Add Component > MyDynamicLink**.
2. Drag **Campfire B** into **Fire**, and your layers into **Burst Layer** and **Noise Layer**.
3. Double-click its Script field. Under `TODO`, **delete the two slashes** in front of one line:

```
float i = fire.CurrentIntensity;   // the fire's intensity: 0 = embers, 1 = blaze
// TODO: YOUR ONE LINE. Delete the two slashes.
burstLayer.burstsPerSecond = Mathf.Lerp(1f, 40f, i);
//   Lerp(1, 40, i): 1 burst a second at embers, about 20 at 0.5, 40 at a blaze.
```

`Mathf.Lerp(1, 40, i)` goes from 1 to 40 as `i`, the fire's Intensity, goes from 0 to 1. This is **RTPC**, real-time parameter control: a value from the game drives a parameter of the sound.

Save, press Play, select Campfire B and drag **Intensity**. Is it more than volume?



Stuck?

- **"Windows found that this file is potentially harmful", or an empty folder after extracting?** Unblock the zip first: right-click it, **Properties**, tick **Unblock**, **OK**, then **Extract all** again.
- **Cannot find the project in Unity Hub?** Skip the Hub: double-click `Assets/W3B_Fire/Scenes/W3B_Student_Shell` in File Explorer.
- **No sound?** Is the layer's Level above 0? The Audio Source volume up? The object ticked?
- **Red error in the Console?** Double-click it: it opens the script at the problem. Check the line ends with `;` and the spelling matches exactly.
- **Changes vanished?** You made them while playing. Make them again when stopped.
- **Recorder trouble, or out of time?** Use the recordings in `Assets/W3B_Fire/Audio/Fire_Recorded`.
- **Too loud?** Start with headphones low: bursts can be sudden.

Ask us. That is what we are here for.



Stretch

Stretch tasks

- **Combine:** one fire with a recorded layer and a procedural layer. Which parts work better each way?
- **Compare:** open `W3B_Complete_Reference` and walk between its two campfires.
- **Interact:** add `AddLogOnKey` to a campfire and press **L** to throw on a log. In your exhibit, `AddLog` could come from a VR hand gesture or interaction instead.
- **Adapt:** on FireController, tick **Burn Down**.
- **More lines:** use both lines in MyDynamicLink, or change the numbers. Can you make the fire sound bigger, not just louder?

At 10:25, be ready to tell us

Bring one fire to play through the speaker, and tell us:

1. What parts did you hear in the mystery sound? What did you call them?
2. Which part worked better recorded, and which procedural?
3. What did each of your lines of code change?

For your coursework: which sound in your exhibits needs to be *dynamic*? Bring it to Monday's session.

Thank you